A one-dimensional MAP subcase for geometric preduals of bounded $C^{0,\omega}$ spaces

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Source and Question

The source paper is Alexander Brudnyi, *On Properties of Geometric Preduals of $C^{k,\omega}$ Spaces*, arXiv:1607.04824. In Section 1.4, after Theorem 1.3, the paper proves that $G_b^{0,\omega}(\mathbb{R}^n)$ has the metric approximation property when $\lim_{t\to\infty}\omega(t) = \infty$. It then observes that in the bounded-modulus case $G_b^{0,\omega}(\mathbb{R}^n)$ is merely isomorphic to a space with the metric approximation property, with distortion depending on ω, and states that it is not clear whether $G_b^{k,\omega}(\mathbb{R}^n)$ itself has the metric approximation property.

This packet proves a scoped positive result for the full-space case $k = 0, n = 1$, under the additional natural hypothesis that the modulus is concave.

Proof Intuition

For $k = 0$, the dual space is the bounded ω-Lipschitz function space: the unit ball consists of functions f with $\|f\|_\infty \le 1$ and $|f(x) - f(y)| \le \omega(|x - y|)$. The key finite-rank approximants come from randomly shifted grids. If $Q_{h,u}(x) = h\lfloor (x + u)/h\rfloor$ and u is uniformly distributed on $[0, h]$, then for $t = mh + s$, $0 \le s < h$,

$$\mathbb{E}\,\omega(|Q_{h,u}(x + t) - Q_{h,u}(x)|) = \left(1 - \frac{s}{h}\right)\omega(mh) + \frac{s}{h}\omega((m + 1)h) \le \omega(t),$$

because ω is concave. Thus averaging over shifted roundings is a contraction for the ω-Lipschitz seminorm. Clipping the rounded points to a large interval makes the range finite-dimensional while preserving contractivity. The mesh goes to zero and the clipping interval expands, so the induced finite-rank contractions on the geometric predual converge to the identity.

Theorem 1. *Let $\omega : [0,\infty) \to [0,\infty)$ be nondecreasing, concave, continuous, and satisfy $\omega(0) = 0$. Then the geometric predual $G_b^{0,\omega}(\mathbb{R})$ of $C_b^{0,\omega}(\mathbb{R})$, in the sense of arXiv:1607.04824, has the metric approximation property.*

Proof. For $k = 0$, the norm on $C_b^{0,\omega}(\mathbb{R})$ is

$$\|f\| = \max\left\{\sup_{x\in\mathbb{R}} |f(x)|,\ \sup_{x\ne y} \frac{|f(x) - f(y)|}{\omega(|x - y|)}\right\}.$$

The geometric predual $G_b^{0,\omega}(\mathbb{R})$ is the closed linear span of the evaluation functionals δ_x inside $C_b^{0,\omega}(\mathbb{R})^*$, and by Brudnyi's Theorem 1.2 we may identify $C_b^{0,\omega}(\mathbb{R})$ with $G_b^{0,\omega}(\mathbb{R})^*$.

Fix $R > 0$ and $h > 0$. Let

$$Q_{h,u}(x) := h \left\lfloor \frac{x+u}{h} \right\rfloor, \qquad 0 \le u < h,$$

and let $C_R(t) = \max\{-R, \min\{t, R\}\}$. Define

$$(T_{R,h}f)(x) := \frac{1}{h} \int_0^h f\big(C_R(Q_{h,u}(x))\big)\, du, \qquad f \in C_b^{0,\omega}(\mathbb{R}).$$

The values $C_R(Q_{h,u}(x))$ belong to the finite set

$$A_{R,h} := \{-R, R\} \cup \{jh : \ -R < jh < R,\ j \in \mathbb{Z}\}.$$

Hence $T_{R,h}$ has finite-dimensional range, since

$$(T_{R,h}f)(x) = \sum_{a \in A_{R,h}} p_a(x) f(a)$$

for suitable scalar coefficient functions p_a.

We next prove $\|T_{R,h}\| \le 1$. The supremum norm is immediate: $\|T_{R,h}f\|_\infty \le \|f\|_\infty$. For the seminorm, take $x < y$ and put $t = y - x$. Write $t = mh + s$, with $m \in \mathbb{Z}_+$ and $0 \le s < h$. As u ranges uniformly over $[0, h]$, the difference $Q_{h,u}(y) - Q_{h,u}(x)$ is mh with probability $1 - s/h$ and $(m+1)h$ with probability s/h. Since C_R is nonexpansive on $\mathbb{R}$ and ω is nondecreasing,

$$\begin{aligned}
\frac{1}{h} \int_0^h \omega(|C_R(Q_{h,u}(y)) - C_R(Q_{h,u}(x))|)\, du &\le \Big(1 - \frac{s}{h}\Big)\, \omega(mh) + \frac{s}{h} \omega((m+1)h) \\
&\le \omega(mh + s) = \omega(|x - y|),
\end{aligned}$$

where the last inequality is concavity of ω. Therefore, for $f \in C_b^{0,\omega}(\mathbb{R})$,

$$\begin{aligned}
|T_{R,h}f(y) - T_{R,h}f(x)| &\le \frac{1}{h} \int_0^h |f(C_R(Q_{h,u}(y))) - f(C_R(Q_{h,u}(x)))|\, du \\
&\le |f|_\omega\, \omega(|x - y|).
\end{aligned}$$

Thus $T_{R,h}$ is a weak-star continuous finite-rank contraction on $C_b^{0,\omega}(\mathbb{R}) = G_b^{0,\omega}(\mathbb{R})^*$. Let $H_{R,h}$ be its preadjoint on $G_b^{0,\omega}(\mathbb{R})$; equivalently,

$$H_{R,h}\delta_x = \frac{1}{h} \int_0^h \delta_{C_R(Q_{h,u}(x))}\, du,$$

which is a convex combination of evaluations at points of $A_{R,h}$. Hence $H_{R,h}$ is a finite-rank contraction.

It remains to check convergence to the identity. Fix $x \in \mathbb{R}$. If $R > |x| + 1$ and $0 < h < 1$, then clipping does not affect $Q_{h,u}(x)$, and $|Q_{h,u}(x) - x| \le h$. Therefore

$$\|H_{R,h}\delta_x - \delta_x\| \le \frac{1}{h} \int_0^h \|\delta_{Q_{h,u}(x)} - \delta_x\|\, du \le \omega(h) \to 0.$$

The linear span of the evaluations δ_x is dense in $G_b^{0,\omega}(\mathbb{R})$, and the operators $H_{R,h}$ are uniformly bounded by one. Taking, for example, $R = N$ and $h = 1/N$, we obtain finite-rank contractions converging strongly to the identity on $G_b^{0,\omega}(\mathbb{R})$, and hence uniformly on compact subsets. This is precisely the metric approximation property. □

Verification Notes and Scope

The proof uses only the $k = 0, n = 1$ structure and concavity of the modulus. It does not solve the higher-dimensional problem, the $k \geq 1$ cases, or the nonconcave moduli allowed by the source paper's weaker hypotheses. For unbounded concave moduli the source already implies the metric approximation property; the new content is the bounded concave-modulus case, where the source only gives an isomorphic MAP comparison and a bounded approximation constant above one.

The shifted-grid contraction is insensitive to boundedness of ω. Examples covered include $\omega(t) = t^\alpha$ for $0 < \alpha \leq 1$ and bounded concave variants such as $\omega(t) = \min\{t^\alpha, 1\}$.

Novelty Check

The run indexes had no exact previous result for arXiv:1607.04824. Searches for the exact arXiv id, for the notation $G_b^{k,\omega}$, for $C_b^{k,\omega}$ and the approximation property, and for snowflake or concave-modulus Lipschitz-free MAP variants did not locate a later direct answer. The local registry has a separate literature-implied packet for arXiv:1501.07036 covering the usual-metric real-line subcase of a Lipschitz-free MAP problem; that does not give this bounded $\omega(|x-y|)$-metric result by itself.

References

- A. Brudnyi, *On Properties of Geometric Preduals of $C^{k,\omega}$ Spaces*, arXiv:1607.04824.
- E. Pernecka and R. J. Smith, *The metric approximation property and Lipschitz-free spaces over subsets of* $\mathbb{R}^N$, arXiv:1501.07036.

Human Review Recommendation

Recommended review focus: verify that the shifted-grid averaging operator is indeed weak-star continuous with the stated preadjoint, and that the concavity inequality is the only extra hypothesis beyond the $k = 0, n = 1$ source setting.